| Learner: Peter-Paul D. Gumpal | Instructor: Professor Roben A. Juanatas |
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Paper Review: The Role of Precision Agriculture in Food Security
Authors: Bruce Erickson, Scott W. Fausti
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Part I – Summary of the Article

Precision Farming (PA), by definition, leverages information technology to manage crop and livestock variability, specifically helping producers who are inundated with complex data to make efficient decisions. The researchers' paper suggests that PA, when implemented as a tool, could benefit global Food Security. However, their study navigates beyond the technical aspects of PA to analyze its socio-economic impact on the four pillars of food security: Availability (production levels), Access (economic ability to buy food), Stability (resilience to disruptions), and Utilization (nutritional quality).

Erickson and Fausti's paper addresses two types of technology adopters based on economic factors: the "Early Adopters" from developed nations like the US and Canada, and the "Laggard Adopters" in the developing world, like the Philippines. Moreover, the core problem identified is "scale bias." This means that PA technologies, such as GNSS guidance (based on satellite navs to guide machineries) and variable rate monitors (or VRTs, such as IoTs) are explicitly designed for large, mechanized farms. Consequently, this leaves smallholder farmers (those with less than two hectares of farmland, comprising the majority of the world's farms) at a competitive disadvantage, potentially risking their livelihoods.

As the research is conceptual in nature, the authors utilized economic analysis and a literature review rather than conducting a field experiment. They synthesized economic theories, such as economies of scale and supply/demand elasticity, with global agricultural statistics from the FAO and USDA. They compared adoption rates across different geographies and farm sizes to project the long-term effects of PA on global food prices and the economic value for farmers.

The paper's key findings apply to both the macro and micro scales. Globally, PA increases food supply and lowers food prices, which benefits consumers (improving Food Access). However, locally, this price drop negatively impacts small farmers who cannot produce higher yields because they cannot adopt the technology. The adoption gap is significant—up to 90% for large

mechanized farms versus near zero for small lands. Finally, the authors note that the value of PA is shifting toward "knowing"—using data to stabilize the supply chain, which is critical during disruptions like the COVID-19 pandemic.

## Part II – Personal Reflection

My ongoing non-profit passion project, "Project TIWALA" (currently in the alpha stage, stands for Temporal and Integrated Weather Analytics for Local Agriculture), is a socio-technical system designed to address farming intelligence democratization gaps. Part of the IT solution is to help agricultural producers who are inundated with complex environmental issues such as Climate change, Agri-political, and financial data to make more efficient farming decisions. More importantly, the social aspect of Project TIWALA focuses on the target beneficiaries—the "smallholder" farmers of the province of Isabela. The goal is to understand what drives them to adopt technology. By leveraging existing theories such as Technology Adoption, Human-Computer Interaction (HCI), and principles like Safety-by-Design, I aim to build a system that fosters "trust" (tiwala), ensuring these essential elements are employed effectively for the project.

In the context of our course, Integrating Diverse Systems and Technology, this paper reframes integration from a technical challenge to a socio-technical one. It argues that integrating technology isn't just about utilizing IoT, connecting it to a cloud server, aggregate the data, and running analytics—it is about fitting that ICT solution into the economic reality of the user. In the case of my project, that user is an underprivileged farmer in the Province of Isabela. The authors note that current PA systems (expensive hardware) fail to integrate with the societal system of smallholder farmers who lack capitalization and mechanization. This creates a digital divide that threatens to leave the poorest agricultural producers behind.

A striking insight from this research is that technology adoption by wealthy farmers can potentially make the poor poorer. If corporate farms use PA to drive down the price of corn or palay, Isabeliños using traditional farming methods will be further disadvantaged. This highlights that a "successful implementation" does not always translate to a win-win situation; the IT and business model must make sense for everyone. However, the paper also introduced the concept of "Non-digital PA," suggesting that precision is a methodology, not just a technology, meaning expensive equipment is optional.

I agree with the authors' diagnosis but find their solutions slightly limited. They correctly identify that "knowledge" is the missing link for small farmers, but they view PA primarily through the lens of mechanization. From a Philippine perspective, I disagree that small size is a permanent

barrier. The Service Provider model (e.g., SaaS), which is central to Project TIWALA, can allow smallholder farmers to access high-tech tools without owning them. The paper touches on this but does not fully explore it as a primary solution for the developing world—perhaps because the authors are not from a developing country like ours.

Regarding the application of these concepts to real-world integration scenarios, specifically in Agricultural Intelligence, we must prioritize accessibility. In this day and age, the Industry 4.0, solution architects such as us DIT students, can simply select the proper AI tools and integrate them seamlessly into the day-to-day farming activities of smallholder producers. By managing the data lifecycle (collection, synthesis, delivery, and validation) and delivering insights through low-tech, high-touch, and readily available tools—such as Facebook and ChatGPT—Precision Agriculture will cease to be a buzzword. Instead, it will become a practical reality, seamlessly integrated for the internet-connected farmer.

**References**


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